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CURRENT OPINION

The Trade Secret Taboo: Open Science Methods are Required to Improve Prediction Models in Sports Medicine and Performance

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Abstract

Clinical prediction models in sports medicine that utilize regression or machine learning techniques have become more widely published, used, and disseminated. However, these models are typically characterized by poor methodology and incomplete reporting, and an inadequate evaluation of performance, leading to unreliable predictions and weak clinical utility within their intended sport population. Before implementation in practice, models require a thorough evaluation. Strong replicable methods and transparency reporting allow practitioners and researchers to make independent judgments as to the model's validity, performance, clinical usefulness, and confidence it will do no harm. However, this is not reflected in the sports medicine literature. As shown in a recent systematic review of models for predicting sports injury models, most were typically characterized by poor methodology, incomplete reporting, and inadequate performance evaluation. Because of constraints imposed by data from individual teams, the development of accurate, reliable, and useful models is highly reliant on external validation. However, a barrier to collaboration is a desire to maintain a competitive advantage; a team's proprietary information is often perceived as high value, and so these 'trade secrets' are frequently guarded. These 'trade secrets' also apply to commercially available models, as developers are unwilling to share proprietary (and potentially profitable) development and validation information. In this Current Opinion, we: (1) argue that open science is essential for improving sport prediction models and (2) critically examine sport prediction models for open science practices.

Key Points

In a recent systematic review of sports injury prediction models, only one of the 30 studies (3%) included in the review made their protocol and analysis plan publicly available.

In a secondary data extraction for this systematic review, none of the 30 studies made their data or code or data publicly available in an open access repository.

Fostering collaboration between sports organizations and unlocking 'trade secrets' through available scientific methods can potentially improve athlete care.

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1 Introduction

In recent years, clinical prediction models in sports medicine that utilize regression or machine learning techniques have become more widely published, used, and disseminated [1]. These models utilize an individual's data (termed predictors herein) to estimate the risk or probability of health-related events being present (diagnosis) or whether they will occur in the future (prognosis), and can be used to facilitate clinical decisions [2]. While models can be used in practice to improve decision making, individualized predictions obtained from poorly developed models are likely to be unreliable. This will impact clinical model utility, but more seriously, making clinical decisions based on inaccurate predictions can even cause patient harm. The accuracy, reliability, and clinical utility of models are highly dependent on the use of robust methods and transparent reporting, and there has been a recent drive to raise awareness of this in the sports medicine literature [1–4].

Before implementation in practice, models require a thorough evaluation as models are typically fit too closely to the

data from which they are developed. As a result, prediction performance declines when tested on unfamiliar populations or people [2]. To account for this, models are assessed and adjusted in a process termed internal validation. Internal validation is defined as adjusting for optimism, using the entire data set through bootstrapping or cross-validation [2]. Additionally, models should be evaluated in relation to the target population (i.e., using populations from an equivalent sport and competition level, or the same population used to develop the model but at a different timepoint). To assess model transportability, performance in populations from different leagues, regions, and sports can be analyzed; a process known as external validation [2]. As practice, training, and clinical strategies change over time, calibration drift can occur even for initially well-performing models [5], such that model predictions become increasingly miscalibrated with actual outcomes. As such, prediction models need periodic monitoring and updating (such as recalibration or re-estimation of the model equation) to reflect the non-stationarity of data owing to changes in both clinical practice and athletic populations [2].

Strong replicable methods and transparent reporting of model development and validation processes are pivotal to all aspects of prediction model research as this allows practitioners and researchers to make independent judgments as to a model's validity, performance, clinical usefulness, and confidence that decisions made from model results will do no harm [6]. However, this is not reflected in the sports medicine literature. As shown in a recent systematic review for sports injury prediction models, most were typically characterized by poor methodology, incomplete reporting, and inadequate performance evaluation [1]. Importantly, just over 1% of models were rated as a low risk of bias, and no studies performed external validation; consequently, there were no models that could be recommended for use in practice [1]. As well as these methodological issues [1], the development and validation of accurate, reliable, and useful models is challenging in practice, often because of sample size constraints imposed by using data from individual teams, although this could be addressed through collaboration and pooling of individual participant data [7]. This issue may restrict external validation processes, which are essential before implementing models in sports medicine practice [7].

However, a barrier to collaboration is a desire to maintain a competitive advantage; a team's proprietary information is often perceived as high value, and so these 'trade secrets' are frequently and carefully guarded [8]. Similar 'trade secrets' also apply to commercially available models, as developers are unwilling to share proprietary (and potentially profitable) development and validation information [3]. Therefore, in this Current Opinion, we: (1) argue that open science is essential for improving sport prediction models and (2) critically examine sport prediction models for open science practices.

2 Implications of Locking in 'Trade Secrets' in Sport and Prediction Models

Current 'trade secret' practices performed in sport have significant implications for athlete health in general and for prediction model development and internal and external validation. It should be noted that this is only one contributor to the lack of open science in sports medicine and prediction research. Other contributors including journal practices and incentives and a lack of open science education may also contribute to these issues. Often, prediction model research is performed within a single club, thus offering only a retrospective account of how specific outcomes occurred for that particular club only during a historical time period. This means that models may be developed on outdated data (so may not account for the effects of athlete transfers or a changing injury incidence for example). A further associated problem is that the limited number of outcome observations available within a single club (injuries for example) means that any developed models are likely to suffer from considerable overfitting, which is common when sample sizes are small [9]. This means that ultimately models are likely to be unreliable and may cause harm when poorly developed models are applied at the club level. This may explain why studies may culminate in different conclusions, despite using similar approaches to modeling. For example, concerning the popular acute chronic workload ratio and its relationship with injury, studies frequently report different results, with many contradicting each other and often finding the completely opposite effect [10].

An essential stage of model development is the identification of important, established, and clinically relevant predictor variables that are associated with a health-related outcome to be predicted [53, 54]. While secondary and exploratory analyses can provide valuable information regarding important predictors, issues with these analyses are common in sport and include 'p-hacking' and hypothesizing after the results are known (HARKing) [11]. 'P-hacking' is defined as performing statistical analyses until a "significant" result is found within the sampled data. 'Harking' is creating hypotheses after analyzing the data, to create 'confirmed' hypotheses [12]. These practices are considered scientific misconduct [12], and can impact bias in reported results [11]. Approximately 30% of scientists in a recent survey reported questionable research practices to assuage results for an improved chance of publication [13].

Open science principles are key to improving sport prediction model performance and examining the impact of these models when implemented in a practical setting

[14]. While complete reporting is an important component, embracing a fuller open science framework means that study registration, study protocols, analysis plans, data, and code are "findable, accessible, interoperable, and reusable (FAIR)" research [15]. An open science environment permits fellow practitioners and researchers to assess, reproduce, and validate models from open data and model code [14, 15]. Open science principles have long been discussed and to varying extent implemented within the physical and biological sciences for many years [16]. However, the physical sciences, such as physics or chemistry, do not have to consider protected patient information [16, 17], a concern in medicine and healthcare. Despite this issue, many medical funding agencies, such as the National Institutes of Health, now require research data and findings to be accessible using open science practices [16–18]. Recently, we described the potential for open science in sport [14]. This would circumvent sample size restrictions associated with data from individual sports teams, small organizations, or leagues, thus creating larger and more current datasets that could lead to developing more impactful models [14]. These large datasets could be used for improved sample sizes for prediction model development [14, 15], recalibration of previously developed models [19], individual patient data meta-analyses [20, 21], or external validation of previously developed models [22]. However, competition between clubs or leagues and the potential for data re-identification are risks in creating an open science environment. Strategies should be employed to ensure basic demographics, exposure, and a minimal set of relevant clinical measures can be shared without sports clubs losing a competitive advantage and protecting athlete identity [14].

3 Current Open Science Practices are Sub-standard for Sport Prediction Models

While we believe open science principles are vital for improving the quality and transparency of sports prediction models, such practices are not widely adopted in sports medicine. In the aforementioned systematic review [1], of the 30 studies that were included, all related to the development of a prediction model and none externally validated a previously developed model. The included studies have also been subsequently assessed for open science practices by the author team. The new data extraction and analyses included establishing protocol availability through an open access database or server, study registration, an indication of using the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) reporting guideline (Appendix 1), [6] data sharing through an open access database or supplement, and availability of the prediction model code (or software objects) to permit implementation (in principle) or evaluation by independent researchers (Table 1). 'Data available upon request' was not considered open access data, as previous research has demonstrated data are not truly available from most authors [23, 24].

Within the new data extraction and analysis results, only one of the 30 (3%) studies reported a study protocol, and the same paper was the only study to register their study [7]. Only three (10%) studies reported adhering to the TRIPOD guidelines for reporting. Only four (13%) studies reported the full model equation, and no machine learning studies provided a link or supplement to model code, software object, or hyperparameter values. Finally, no studies deposited the data in an open access database or shared the data, although four studies used open access datasets available from the Internet [1].

4 Suggestions to Improve Open Science for Sport Prediction Models

Many of the highlighted issues with prediction model development and validation can be remediated via adopting open science principles in sport. General information can be provided without divulging specific details of training, recovery, or rehabilitation interventions (including examples such as injury outcomes, distances covered at various speeds, running speeds, the number of accelerations and decelerations completed, training time and results from physical performance, and anthropometric tests) in the form of de-identified datasets. Any software code created during model development or validation could also be shared. The most widely used open code repository is GitHub (https://github.com/). Introducing these changes would allow other researchers and practitioners to combine the findings from other studies with their own in an effort to build on prior knowledge and increase sample size sufficiently to allow the conduct of adequately powered studies and create better models for the club that originally developed the model and the league as a whole.

Publishing a priori research and analytical plan registration protocols can further improve transparency [11]. Further, such an approach allows others to review the proposed methodology, check code, explore model assumptions, and try other types of data analysis to address different research questions. Examples of open repositories include Open Science Framework (https://osf.io/) and Zenodo (https://zenodo.org/). Collectively, open science allows practitioners working within different levels of competition and sports to share ideas, learn from each other, disseminate findings, and advance the knowledge of athlete healthcare at a faster pace than an isolated research group would conducting small sample studies within a single club.

Table 1 Systematic review [1] included studies and open science reporting

Study	Year	Protocol availability	Study registration	Use of TRIPOD guideline	Data sharing	Availability of full prediction model equation, code, or logic
Abalo-Nunez et al. [25]	2018	No	No	No	No	Yes
Ayala et al. [26]	2019	No	No	No	No	No
Carbuhn et al. [27]	2020	No	No	No	No	Yes
Carey et al. [28]	2018	No	No	No	No	No
Colby et al. [29]	2018	No	No	No	No	No
Feijen et al. [30]	2021	No	No	Yes	No	No
Gabbet [31]	2010	No	No	No	No	No
Hughes et al. [7]	2020	Yes: https://doi.org/10.1186/s41512-019-0063-8	Yes: clinicaltrials.gov: NCT03782389	Yes	No	Yes
Ivarsson et al. [32]	2014	No	No	No	No	No
Jauhiainen et al. [33]	2020	No	No	No	No	No
Karuc et al. [34]	2021	No	No	Yes	No	No
Khayambashi et al. [35]	2015	No	No	No	No	No
Landis et al. [36]	2018	No	No	No	No	No
Lopez-Valencia et al. [37]	2018	No	No	No	No	No
Luu et al. [38]	2020	No	No	No	No	No
McCann et al. [39]	2018	No	No	No	No	No
Oliver et al. [40]	2020	No	No	No	No	No
Pontillo et al. [41]	2021	No	No	No	No	No
Powers et al. [42]	2018	No	No	No	No	No
Rommers et al. [43]	2020	No	No	No	No	No
Rossi et al. [44]	2018	No	No	No	No	No
Ruddy et al. [45]	2018	No	No	No	No	No
Shambaugh et al. [46]	1991	No	No	No	No	Yes
Sturnick et al. [47]	2015	No	No	No	No	No
Teramoto et al. [48]	2017	No	No	No	No	No
Thornton et al. [49]	2017	No	No	No	No	No
van der Does et al. [50]	2015	No	No	No	No	No
Whiteside et al. [51]	2016	No	No	No	No	No
Wiese et al. [52]	2014	No	No	No	No	No
Wilkerson et al. [53]	2015	No	No	No	No	No

TRIPOD Transparent Reporting of multivariable prediction model for Individual Prognosis Or Diagnosis [6]

Nevertheless, despite the advantages of open science there are certain issues that need consideration. With the unique competition and athlete requirements in sport, de-identified data may not be suitable in a pure open access to all format. In particular, to protect athlete re-identification, using independent gate keeper data warehouses (managed by sport leagues or governing bodies) can facilitate anonymity, and may be pertinent [16, 17]. Gate keeper data warehouses are a centralized independent organization that both manage data collected and stored on their servers, and utilize committees who review (and then accept or deny) requests for data usage for study purposes [17]. While some gate keeper warehouses already exist in sport [54], they only provide injury data, without other medical biomarkers such as range of motion, strength, or training load, decreasing their usability.

Aside from gatekeeper warehouses, another potential solution is for each club or organization to be required to provide a synthetic (also known as simulated) data set each season. A synthetic data set contains data that are artificially developed, based on real data [55]. While this would increase the medical and analytical burden, the risk of athlete re-identification and releasing protected health information is more mitigated [55, 56].

5 Conclusions

Currently published sport prediction models are poorly developed and reported in general, and none are viable for use [1]. This is compounded because of sample size limitations associated with conducting such studies on small populations (such as individual sports teams) and the lack of open science and availability of data and code, which prevents future model improvements. Fostering collaboration between sports organizations and unlocking 'trade secrets' through available scientific methods can potentially leverage data from multiple teams or leagues, facilitating the development, validation, and implementation of better and more impactful models across broader populations that will benefit all teams and organizations. This has significantly greater potential to improve athlete care than current practice. Open science can also facilitate scientific collaboration and quality improvement, through non-partial third-party evaluation, including error evaluations, assessing data structure, and model interpretation [14]. However, until open science practices are fully adopted in sports, we will continue to make avoidable mistakes and produce poor-quality clinical prediction models, which is likely to impede medical progress over time.

Appendix 1: TRIPOD Checklist: Prediction Model Development and Validation

Section/topic	Item		Checklist item	Page
Title and abstract				
Title	1	D;V	Identify the study as developing and/or validating a multivariable prediction model, the target population, and the outcome to be predicted	
Abstract	2	D;V	Provide a summary of objectives, study design, setting, participants, sample size, predictors, outcome, statistical analysis, results, and conclusions	
Introduction				
Background and objectives	3a	D;V	Explain the medical context (including whether diagnostic or prognostic) and rationale for developing or validating the multivariable prediction model, including references to existing models	
	3b	D;V	Specify the objectives, including whether the study describes the development or validation of the model or both	
Methods				
Source of data	4a	D;V	Describe the study design or source of data (e.g., randomized trial, cohort, or registry data), separately for the development and validation data sets, if applicable	
	4b	D;V	Specify the key study dates, including start of accrual; end of accrual; and, if applicable, end of follow-up	
Participants	5a	D;V	Specify key elements of the study setting (e.g., primary care, secondary care, general population) including number and location of centers	
	5b	D;V	Describe eligibility criteria for participants	
	5c	D;V	Give details of treatments received, if relevant	
Outcome	6a	D;V	Clearly define the outcome that is predicted by the prediction model, including how and when assessed	
	6b	D;V	Report any actions to blind assessment of the outcome to be predicted	
Predictors	7a	D;V	Clearly define all predictors used in developing or validating the multivariable prediction model, including how and when they were measured	
	7b	D;V	Report any actions to blind assessment of predictors for the outcome and other predictors	
Sample size	8	D;V	Explain how the study size was arrived at	

Section/topic	Item		Checklist item	Page
Missing data	9	D;V	Describe how missing data were handled (e.g., complete-case analysis, single imputation, multiple imputation) with details of any imputation method	
Statistical analysis methods	10a	D	Describe how predictors were handled in the analyses	
	10b	D	Specify type of model, all model-building procedures (including any predictor selection), and method for internal validation	
	10c	V	For validation, describe how the predictions were calculated	
	10d	D;V	Specify all measures used to assess model performance and, if relevant, to compare multiple models	
	10e	V	Describe any model updating (e.g., recalibration) arising from the validation, if done	
Risk groups	11	D;V	Provide details on how risk groups were created, if done	
Development vs validation	12	V	For validation, identify any differences from the development data in setting, eligibility criteria, outcome, and predictors	
Results				
Participants	13a	D;V	Describe the flow of participants through the study, including the number of participants with and without the outcome and, if applicable, a summary of the follow-up time. A diagram may be helpful	
	13b	D;V	Describe the characteristics of the participants (basic demographics, clinical features, available predictors), including the number of participants with missing data for predictors and outcome	
	13c	V	For validation, show a comparison with the development data of the distribution of important variables (demographics, predictors and outcome)	
Model development	14a	D	Specify the number of participants and outcome events in each analysis	
	14b	D	If done, report the unadjusted association between each candidate predictor and outcome	
Model specification	15a	D	Present the full prediction model to allow predictions for individuals (i.e., all regression coefficients, and model intercept or baseline survival at a given time point)	
	15b	D	Explain how to the use the prediction model	
Model performance	16	D;V	Report performance measures (with CIs) for the prediction model	
Model updating	17	V	If done, report the results from any model updating (i.e., model specification, model performance)	
Discussion				
Limitations	18	D;V	Discuss any limitations of the study (such as nonrepresentative sample, few events per predictor, missing data)	
Interpretation	19a	V	For validation, discuss the results with reference to performance in the development data, and any other validation data	
	19b	D;V	Give an overall interpretation of the results, considering objectives, limitations, results from similar studies, and other relevant evidence	
Implications	20	D;V	Discuss the potential clinical use of the model and implications for future research	
Other information				
Supplementary information	21	D;V	Provide information about the availability of supplementary resources, such as study protocol, Web calculator, and data sets	
Funding	22	D;V	Give the source of funding and the role of the funders for the present study	

CIs confidence intervals, *TRIPOD* Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis

Items relevant only to the development of a prediction model are denoted by D, items relating solely to a validation of a prediction model are denoted by V, and items relating to both are denoted D;V. We recommend using the TRIPOD Checklist in conjunction with the TRIPOD Explanation and Elaboration document

Declarations

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Conflict of interest Garrett S. Bullock, Patrick Ward, Franco M. Impellizzeri, Tom Hughes, Paula Dhiman, Richard D. Riley, and Gary S. Collins have no conflicts of interest that are directly relevant to the content of this article.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Availability of data and material Not applicable.

Code availability Not applicable.

Authors' contributions GB, PW, and GSC conceived the study idea. GB, PW, FMI, TH, RR, SK, and GC were involved in the design and planning. GB and GSC wrote the first draft of the manuscript. GB, PW, FMI, SK, TH, PD, RR, and GSC critically revised the manuscript. All authors approved the final version of the manuscript.

References

1. Bullock GS, Mylott J, Hughes T, Nicholson KF, Riley RD, Collins GS. Just how confident can we be in predicting sports injuries? A systematic review of the methodological conduct and performance of existing musculoskeletal injury prediction models in sport. Sport Med. 2022;42(1):2469–82.
2. Bullock GS, Hughes T, Sergeant JC, Callaghan MJ, Riley RD, Collins GS. Clinical prediction models in sports medicine: a guide for clinicians and researchers. J Orthop Sport Phys Ther. 2021;51(10):517–25.
3. Bullock GS, Hughes T, Arundale AH, Ward P, Collins GS, Kluzek S. Black box prediction methods in sports medicine deserve a red card for reckless practice: a change of tactics is needed to advance athlete care. Sport Med. 2022;52(8):2799–801.
4. Bullock GS, Hughes T, Sergeant JC, Callaghan MJ, Riley R, Collins G. Methods matter: clinical prediction models will benefit sports medicine practice, but only if they are properly developed and validated. Br J Sport Med. 2021;22:1319–21.
5. Davis SE, Lasko TA, Chen G, Siew ED, Matheny ME. Calibration drift in regression and machine learning models for acute kidney injury. J Am Med Inform Assoc. 2017;24(6):1052–61.
6. Collins GS, Reitsma JB, Altman DG, Moons KG. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. J Br Surg. 2015;102(3):148–58.
7. Hughes T, Riley RD, Callaghan MJ, Sergeant JC. The value of Preseason screening for injury prediction: the development and internal validation of a multivariable prognostic model to predict indirect muscle injury risk in elite football (soccer) players. Sport Med Open. 2020;6(1):1–13.
8. Ford RA. Trade secrets and information security in the age of sports analytics. Oxford: The Oxford Handbook of American Sports Law; 2018.
9. Bahr R, Holme I. Risk factors for sports injuries: a methodological approach. Br J Sport Med. 2003;37(5):384–92.
10. Impellizzeri FM, Ward P, Coutts AJ, Bornn L, McCall A. Training load and injury part 2: questionable research practices hijack the truth and mislead well-intentioned clinicians. J Orthop Sport Phys Ther. 2020;50(10):577–84.
11. Caldwell AR, Vigotsky AD, Tenan MS, Radel R, Mellor DT, Kreutzer A, et al. Moving sport and exercise science forward: a call for the adoption of more transparent research practices. Sport Med. 2020;50(3):449–59.
12. Andrade C. HARKing, cherry-picking, p-hacking, fishing expeditions, and data dredging and mining as questionable research practices. J Clin Psychiatry. 2021;82(1):25941.
13. John LK, Loewenstein G, Prelec D. Measuring the prevalence of questionable research practices with incentives for truth telling. Psych Sci. 2012;23(5):524–32.
14. Bullock GS, Ward P, Peters S, Arundale AJH, Murray A, Impellizzeri FM, et al. Call for open science in sports medicine. Br J Sport Med. 2022;56:105719.
15. Wilkinson MD, Dumontier M, Aalbersberg IJ, Appleton G, Axton M, Baak A, et al. The FAIR guiding principles for scientific data management and stewardship. Sci Data. 2016;3(1):1–9.
16. Rockhold F, Nisen P, Freeman A. Data sharing at a crossroads. N Engl J Med. 2016;375(12):1115–7.
17. Bertagnolli MM, Sartor O, Chabner BA, Rothenberg ML, Khozin S, Hugh-Jones C, et al. Advantages of a truly open-access data-sharing model. N Eng J Med. 2017;376(12):1178–81.
18. Kadakia KT, Beckman AL, Ross JS, Krumholz HM. Leveraging open science to accelerate research. N Engl J Med. 2021;384(17): e61.
19. Janssen K, Moons K, Kalkman C, Grobbee D, Vergouwe Y. Updating methods improved the performance of a clinical prediction model in new patients. J Clin Epidemiol. 2008;61(1):76–86.
20. Riley RD, Ensor J, Snell KI, Debray TP, Altman DG, Moons KG, et al. External validation of clinical prediction models using big datasets from e-health records or IPD meta-analysis: opportunities and challenges. BMJ. 2016;353: i3140.
21. Riley RD, Lambert PC, Abo-Zaid G. Meta-analysis of individual participant data: rationale, conduct, and reporting. BMJ. 2010;340: c221.
22. Bleeker S, Moll H, Steyerberg EA, Donders A, Derksen-Lubsen G, Grobbee D, et al. External validation is necessary in prediction research: a clinical example. J Clin Epidemiol. 2003;56(9):826–32.
23. Tedersoo L, Küngas R, Oras E, Köster K, Eenmaa H, Leijen Ä, et al. Data sharing practices and data availability upon request differ across scientific disciplines. Sci Data. 2021;8(1):1–11.
24. Krawczyk M, Reuben E. (Un) available upon request: field experiment on researchers' willingness to share supplementary materials. Acc Res. 2012;19(3):175–86.
25. Abalo-Núñez R, Gutiérrez-Sánchez A, Pérez MI, Vernetta-Santana M. Injury prediction in aerobic gymnastics based on anthropometric variables. Sci Sport. 2018;33(4):228–36.
26. Ayala F, López-Valenciano A, Martín JAG, Croix MDS, Vera-Garcia FJ, del Pilar G-V, et al. A preventive model for hamstring injuries in professional soccer: learning algorithms. Int J Sport Med. 2019;40(05):344–53.
27. Carbuhn AF, Sanchez Z, Fry AC, Reynolds MR, Magee LM. A simplified prediction model for lower extremity long bone stress injuries in male endurance running athletes. Clin J Sport Med. 2020;30(5):e124–6.

28. Carey DL, Crossley KM, Whiteley R, Mosler A, Ong K-L, Crow J, et al. Modeling training loads and injuries: the dangers of discretization. Med Sci Sport Exerc. 2018;50(11):2267–76.
29. Colby MJ, Dawson B, Peeling P, Heasman J, Rogalski B, Drew MK, et al. Improvement of prediction of noncontact injury in elite Australian footballers with repeated exposure to established high-risk workload scenarios. Int J Sport Physiol Perform. 2018;13(9):1130–5.
30. Feijen S, Struyf T, Kuppens K, Tate A, Struyf F. Prediction of shoulder pain in youth competitive swimmers: the development and internal validation of a prognostic prediction model. Am J Sport Med. 2021;49(1):154–61.
31. Gabbett TJ. The development and application of an injury prediction model for noncontact, soft-tissue injuries in elite collision sport athletes. J Strength Cond Res. 2010;24(10):2593–603.
32. Ivarsson A, Johnson U, Lindwall M, Gustafsson H, Altemyr M. Psychosocial stress as a predictor of injury in elite junior soccer: a latent growth curve analysis. J Sci Med Sport. 2014;17(4):366–70.
33. Jauhiainen S, Kauppi J-P, Leppänen M, Pasanen K, Parkkari J, Vasankari T, et al. New machine learning approach for detection of injury risk factors in young team sport athletes. Int J Sport Med. 2021;42(02):175–82.
34. Karuc J, Mišigoj-Durakovic M, Šarlija M, Markovic G, Hadžic V, Trošt-Bobic T, et al. Can injuries be predicted by functional movement screen in adolescents? The application of machine learning. J Strength Cond Res. 2021;35(4):910–9.
35. Khayambashi K, Ghoddosi N, Straub RK, Powers CM. Hip muscle strength predicts noncontact anterior cruciate ligament injury in male and female athletes: a prospective study. Am J Sport Med. 2016;44(2):355–61.
36. Landis SE, Baker RT, Seegmiller JG. Non-contact anterior cruciate ligament and lower extremity injury risk prediction using functional movement screen and knee abduction moment: an epidemiological observation of female intercollegiate athletes. Int J Sport Phys Ther. 2018;13(6):973.
37. López-Valenciano A, Ayala F, Puerta JM, Croix MDS, Vera-García F, Hernández-Sánchez S, et al. A preventive model for muscle injuries: a novel approach based on learning algorithms. Med Sci Sport Exerc. 2018;50(5):915.
38. Luu BC, Wright AL, Haeberle HS, Karnuta JM, Schickendantz MS, Makhni EC, et al. Machine learning outperforms logistic regression analysis to predict next-season NHL player injury: an analysis of 2322 players from 2007 to 2017. Orthop J Sport Med. 2020;8(9):2325967120953404.
39. McCann RS, Kosik KB, Terada M, Beard MQ, Buskirk GE, Gribble PA. Acute lateral ankle sprain prediction in collegiate women's soccer players. Int J Sport Phys Ther. 2018;13(1):12.
40. Oliver JL, Ayala F, Croix MBDS, Lloyd RS, Myer GD, Read PJ. Using machine learning to improve our understanding of injury risk and prediction in elite male youth football players. J Sci Med Sport. 2020;23(11):1044–8.
41. Pontillo M, Hines SM, Sennett BJ. Prediction of ACL injuries from vertical jump kinetics in Division 1 collegiate athletes. Int J Sport Phys Ther. 2021;16(1):156.
42. Powers CM, Ghoddosi N, Straub RK, Khayambashi K. Hip strength as a predictor of ankle sprains in male soccer players: a prospective study. J Athl Train. 2017;52(11):1048–55.
43. Rommers N, Rössler R, Verhagen E, Vandecasteele F, Verstockt S, Vaeyens R, et al. A machine learning approach to assess injury risk in elite youth football players. Med Sci Sport Exerc. 2020;52(8):1745–51.
44. Rossi A, Pappalardo L, Cintia P, Iaia FM, Fernández J, Medina D. Effective injury forecasting in soccer with GPS training data and machine learning. PLoS One. 2018;13(7): e0201264.
45. Ruddy J, Shield A, Maniar N, Williams M, Duhig S, Timmins R, et al. Predictive modeling of hamstring strain injuries in elite Australian footballers. Med Sci Sport Exerc. 2018;50(5):906–14.
46. Shambaugh JP, Klein A, Herbert JH. Structural measures as predictors of injury basketball players. Med Sci Sport Exerc. 1991;23(5):522–7.
47. Sturnick DR, Vacek PM, DeSarno MJ, Gardner-Morse MG, Tourville TW, Slauterbeck JR, et al. Combined anatomic factors predicting risk of anterior cruciate ligament injury for males and females. Am J Sport Med. 2015;43(4):839–47.
48. Teramoto M, Cross CL, Cushman DM, Maak TG, Petron DJ, Willick SE. Game injuries in relation to game schedules in the National Basketball Association. J Sci Med Sport. 2017;20(3):230–5.
49. Thornton HR, Delaney JA, Duthie GM, Dascombe BJ. Importance of various training-load measures in injury incidence of professional rugby league athletes. Int J Sport Physiol Perform. 2017;12(6):819–24.
50. Van Der Does H, Brink M, Benjaminse A, Visscher C, Lemmink K. Jump landing characteristics predict lower extremity injuries in indoor team sports. Int J Sport Med. 2016;37(03):251–6.
51. Whiteside D, Martini DN, Lepley AS, Zernicke RF, Goulet GC. Predictors of ulnar collateral ligament reconstruction in Major League Baseball pitchers. Am J Sport Med. 2016;44(9):2202–9.
52. Wiese BW, Boone JK, Mattacola CG, McKeon PO, Uhl TL. Determination of the functional movement screen to predict musculoskeletal injury in intercollegiate athletics. Athl Train Sport Health Care. 2014;6(4):161–9.
53. Wilkerson GB, Colston MA. A refined prediction model for core and lower extremity sprains and strains among collegiate football players. J AThl Train. 2015;50(6):643–50.
54. Pollack KM, D'Angelo J, Green G, Conte S, Fealy S, Marinak C, et al. Developing and implementing Major League Baseball's health and injury tracking system. Am J Epidemiol. 2016;183(5):490–6.
55. Snoke J, Raab GM, Nowok B, Dibben C, Slavkovic A. General and specific utility measures for synthetic data. J R Stat Soc A. 2018;181(3):663–88.
56. Abowd JM, Vilhuber L. How protective are synthetic data? International Conference on Privacy in Statistical Databases; 2008. p. 239–46.

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